R statistical analysis

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```r
library("dplyr")
library("tidyr")
library("ggplot2")
library("Hmisc")          # rcorr
library("ppcor")          # pcor
library("GGally")     # ggpairs
library("ggcorrplot") # ggcorrplot
library("gridExtra")  # grid.arrange
library("ggplot2")
library("agricolae")
library("car")
library("nortest")

library(ggcorrplot)
```

TASK 1

```r
# Load data
ChildCarSeats <- read.csv("data/ChildCarSeats.csv")

head(dt)
```

```
##
## 1 function (x, df, ncp, log = FALSE)
## 2 {
## 3     if (missing(ncp))
## 4         .Call(C_dt, x, df, log)
## 5     else .Call(C_dnt, x, df, ncp, log)
## 6 }
```

```r
# Oneway ANOVA
anova_model <- aov(Sales ~ Advert, data = ChildCarSeats)

summary(anova_model)
```

```
##               Df   Sum Sq Mean Sq F value Pr(>F)
## Advert         2 10095311 5047656   92.23 <2e-16 ***
## Residuals    397 21727436   54729
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
cat("P-value is 2e-16, which is very small, so we reject H0.")
```

```
## P-value is 2e-16, which is very small, so we reject H0.
```

```r
cat("Conclusion: advertisements have a statistically significant effect on sales.")
```

```
## Conclusion: advertisements have a statistically significant effect on sales.
```

```r
leveneTest(Sales ~ Advert, data = ChildCarSeats)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   2  0.8088 0.4461
##       397
```

```r
cat("P-value is 0.44 (> 0.05), variances are homogeneous (ANOVA assumption satisfied).")
```

```
## P-value is 0.44 (> 0.05), variances are homogeneous (ANOVA assumption satisfied).
```

```r
residuals <- residuals(anova_model)
shapiro.test(residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals
## W = 0.99583, p-value = 0.3708
```

```r
cat("P-value is 0.37 (> 0.05), residuals follow a normal distribution (ANOVA assumption satisfied).")
```

```
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```
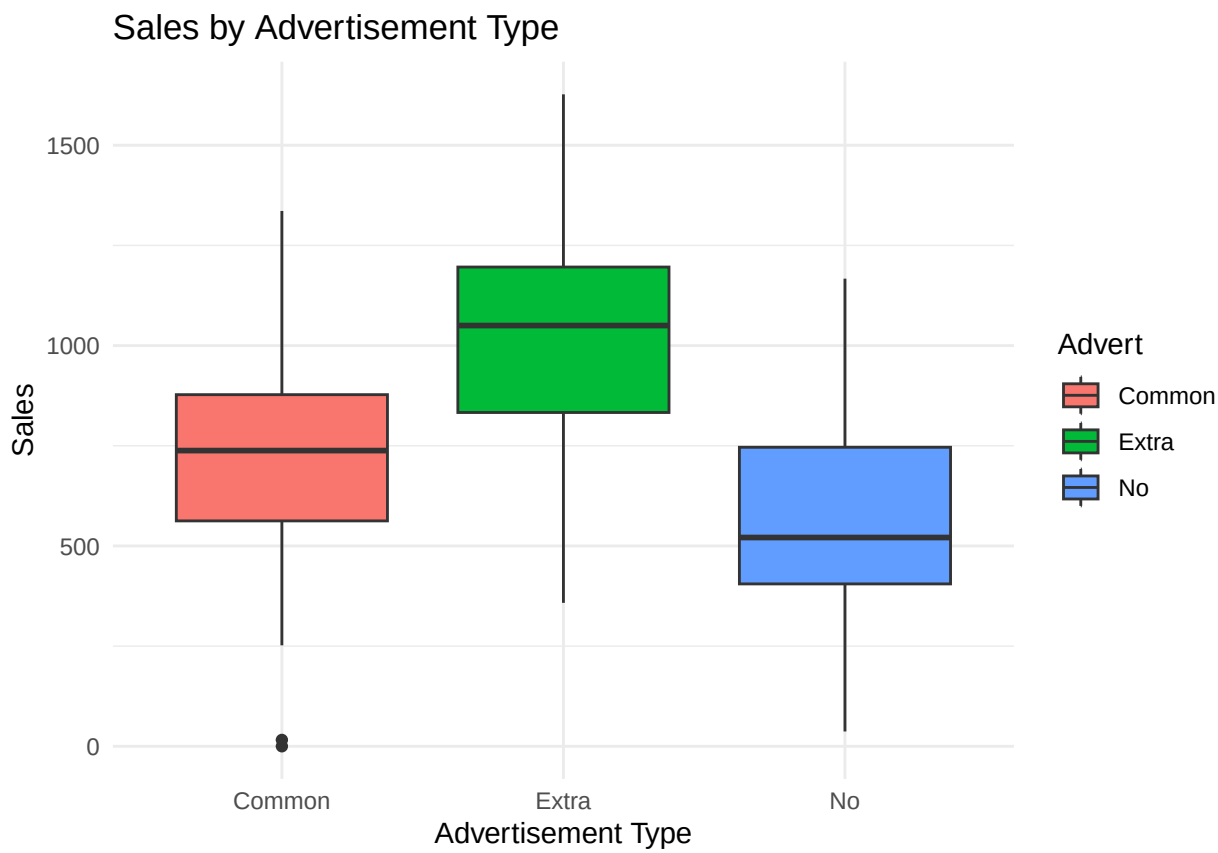

```r
# Boxplot for visualization
ggplot(ChildCarSeats, aes(x = Advert, y = Sales, fill = Advert)) +
  geom_boxplot() +
  labs(title = "Sales by Advertisement Type", y = "Sales", x = "Advertisement Type") +
  theme_minimal()
```




```r
# Tukey's HSD test
tukey_results <- TukeyHSD(anova_model)
print(tukey_results)
```

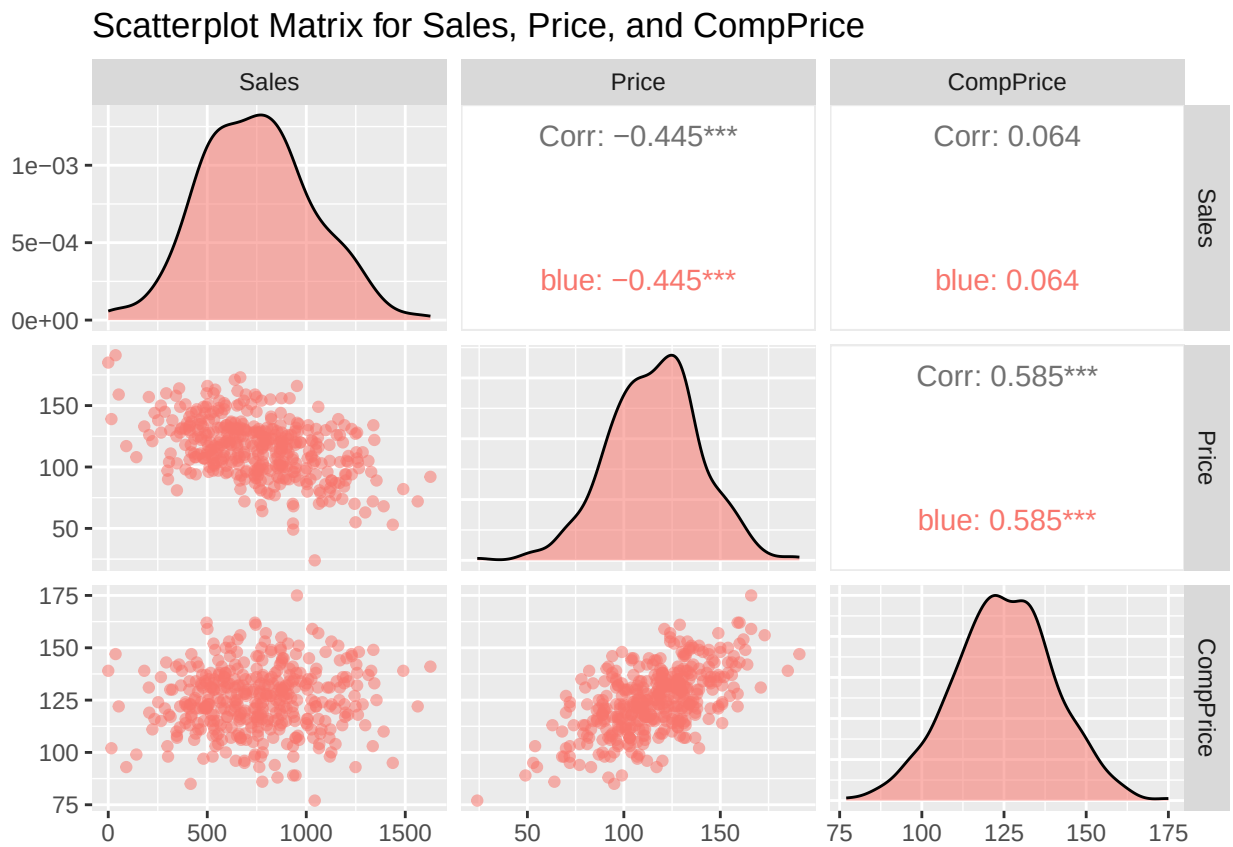
```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = Sales ~ Advert, data = ChildCarSeats)
##
## $Advert
##                   diff       lwr       upr p adj
## Extra-Common  290.7425  220.4108  361.0741     0
## No-Common    -178.3659 -245.7322 -110.9995     0
## No-Extra     -469.1083 -551.0754 -387.1412     0
```

```r
cat("
Extra advertisement leads to significantly higher sales than Common (mean difference: 290.74).
No advertisement results in significantly lower sales than Common (-178.37).
No advertisement yields the lowest sales, with a large gap compared to Extra (-469.11).")
```

```
##
## Extra advertisement leads to significantly higher sales than Common (mean difference: 290.74).
## No advertisement results in significantly lower sales than Common (-178.37).
## No advertisement yields the lowest sales, with a large gap compared to Extra (-469.11).
```

TASK 2

```r
# Scatterplot matrix for Sales, Price, CompPrice
ggpairs(ChildCarSeats[, c("Sales", "Price", "CompPrice")],
        title = "Scatterplot Matrix for Sales, Price, and CompPrice",
        aes(color = "blue", alpha = 0.5))
```




```r
# Calculate Pearson correlation matrix
cor_matrix <- cor(ChildCarSeats[, c("Sales", "Price", "CompPrice")], method = "pearson")
print(cor_matrix)
```

```
##                 Sales      Price  CompPrice
## Sales      1.00000000 -0.4449507 0.06407873
## Price     -0.44495073  1.0000000 0.58484777
## CompPrice  0.06407873  0.5848478 1.00000000
```

```r
# Plot correlation matrix
plot11 <- ggcorrplot(cor_matrix,
                     title = "Pearson's Correlations",
                     lab = TRUE,
                     type = "full",
                     method = "square",
                     outline.color = "white",
```

```r
                    show.legend = FALSE)
print(plot11)
```

```r
# Values closer to 1 indicate a strong positive relationship, -1 a strong negative one, and 0 no relati
```

```r
# Calculate variance-covariance matrix
var_cov_matrix <- cov(ChildCarSeats[, c("Sales", "Price", "CompPrice")])
print(var_cov_matrix)
```

```
##                Sales      Price CompPrice
## Sales     79756.2581 -2975.1908  277.5021
## Price     -2975.1908   560.5844  212.3407
## CompPrice   277.5021   212.3407  235.1472
```

```r
cat("Covariance indicates that Price and Sales are inversely proportional, while Sales and CompPrice ha
```

```
## Covariance indicates that Price and Sales are inversely proportional, while Sales and CompPrice have
```

```r
# Calculate sample standard deviations
sample_sd <- sqrt(diag(var_cov_matrix))
names(sample_sd) <- c("Sales", "Price", "CompPrice")
print(sample_sd)
```

```
##     Sales     Price CompPrice
## 282.41150  23.67666  15.33451
```

```r
cat("Sales shows the highest variability, while Price and CompPrice have low variability.")
```

```
## Sales shows the highest variability, while Price and CompPrice have low variability.
```

TASK 3

```r
ChildCarSeats <- read.csv("data/ChildCarSeats.csv")

head(ChildCarSeats)
```

```
##   Sales CompPrice Price Advert City Expensive
## 1  1122       111    83  Extra  Yes        No
## 2  1185       136   120  Extra  Yes        No
## 3   654       132   124 Common   No       Yes
## 4   901       121   100     No   No        No
## 5  1229       147   131  Extra  Yes        No
## 6   587       121   109 Common  Yes        No
```

```r
# Contingency table
table_expensive_city <- table(ChildCarSeats$Expensive, ChildCarSeats$City)
print(table_expensive_city)
```

```
##
##        No Yes
##   No   58 140
##   Yes  60 142
```

```r
# Calculate odds ratio
odds_ratio <- (table_expensive_city[2, 2] / table_expensive_city[1, 2]) /
              (table_expensive_city[2, 1] / table_expensive_city[1, 1])
print(paste("odds ratio: ", odds_ratio))
```

```
## [1] "odds ratio:  0.98047619047619"
```

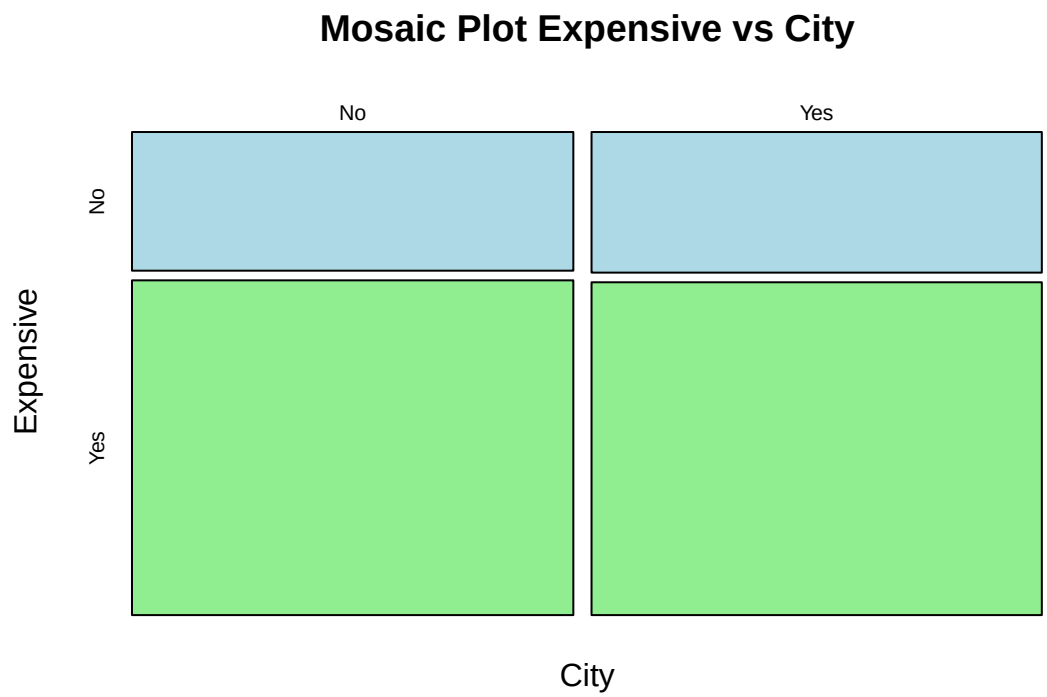
```r
# Calculate the standard error for the log(odds ratio)
se_log_or <- sqrt(1/table_expensive_city[1, 1] + 1/table_expensive_city[2, 1] +
                  1/table_expensive_city[1, 2] + 1/table_expensive_city[2, 2])

# Calculate the log of the odds ratio
log_or <- log(odds_ratio)
conf_int_log_or <- c(log_or - 1.96 * se_log_or, log_or + 1.96 * se_log_or)
conf_int_or <- exp(conf_int_log_or)
print(paste("95% Confidence Interval odds ratio: [", conf_int_or[1], ", ", conf_int_or[2], "]", sep = "
```

```
## [1] "95% Confidence Interval odds ratio: [0.637916491203568, 1.50698966611905]"
```

```r
cat("The CI [0.6379, 1.5069] spans 1, meaning there is no statistically significant association between
```

```
## The CI [0.6379, 1.5069] spans 1, meaning there is no statistically significant association between C
```

```r
# Mosaic plot for Expensive vs City
mosaicplot(table_expensive_city, main = "Mosaic Plot Expensive vs City",
           color = c("lightblue", "lightgreen"),
           xlab = "City", ylab = "Expensive")
```



TASK 4

```r
ChildCarSeats <- read.csv("data/ChildCarSeats.csv")

# Build linear regression model
model <- lm(Sales ~ Price + City, data = ChildCarSeats)
summary(model)
```

```
##
## Call:
## lm(formula = Sales ~ Price + City, data = ChildCarSeats)
##
```

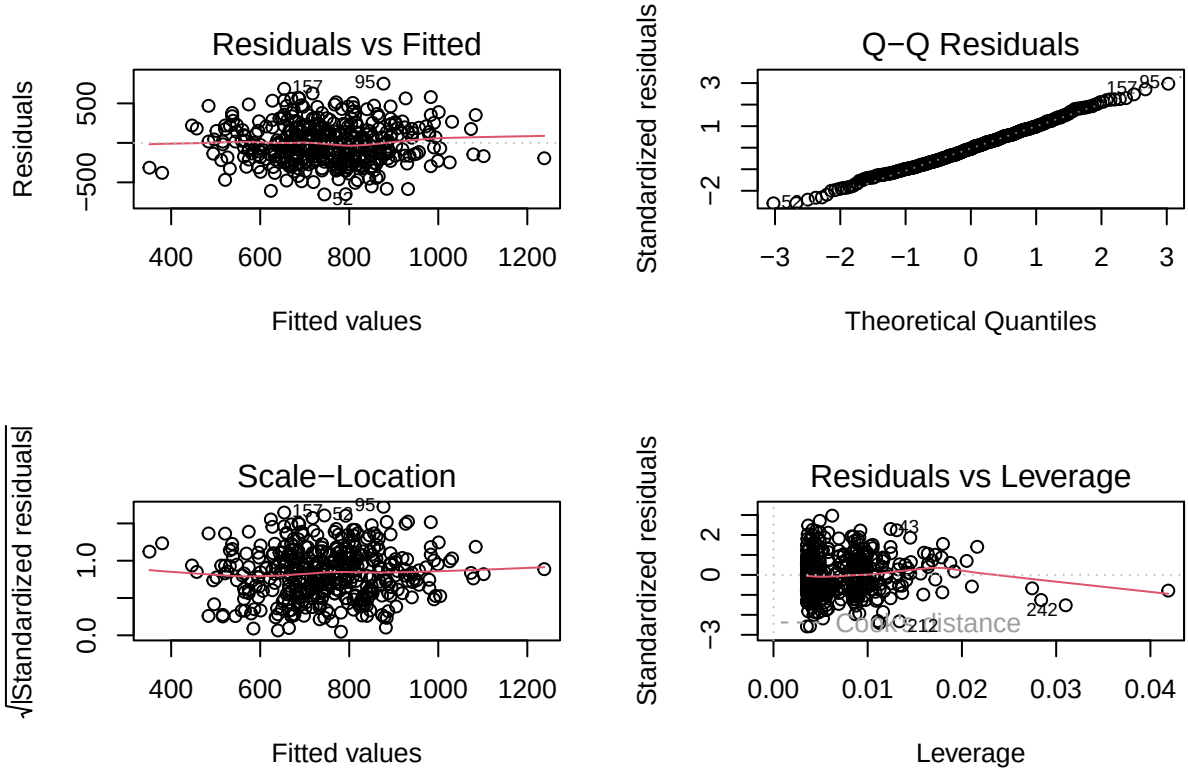
```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -653.24 -184.41  -14.43  166.62  750.00
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1362.1458    65.5230  20.789   <2e-16 ***
## Price         -5.3104     0.5367  -9.895   <2e-16 ***
## CityYes        3.4095    27.8293   0.123    0.903
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 253.5 on 397 degrees of freedom
## Multiple R-squared:  0.198,  Adjusted R-squared:  0.194
## F-statistic: 49.01 on 2 and 397 DF,  p-value: < 2.2e-16
```

```r
cat("
The model shows that Price has a significant negative effect on Sales (as price increases, sales decrea
City does not have a significant effect on Sales (high p-value: 0.903).
The model is statistically significant but explains only ~19.4% of the variability in sales (R-squared)
```

```
##
## The model shows that Price has a significant negative effect on Sales (as price increases, sales dec
## City does not have a significant effect on Sales (high p-value: 0.903).
## The model is statistically significant but explains only ~19.4% of the variability in sales (R-squar
```

```r
# Plot the model
ggplot(ChildCarSeats, aes(x = Price, y = Sales, color = City)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE, color = "blue") +
  labs(title = "Regression of Sales on Price and City",
       x = "Price", y = "Sales") +
  theme_minimal()
```

```
par(mfrow = c(2, 2))
plot(model)
```




```
cat("
Residuals vs Fitted: ideally should show a random scatter (residuals not systematically related to fitte

QQ Residuals: if residuals are normally distributed, points should roughly follow the diagonal line.

Scale-Location: random scatter with no clear pattern (equal variance of residuals across all levels of p

Residuals vs Leverage: highlights points that are outliers or have high leverage."
)
```

```
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## Residuals vs Fitted: ideally should show a random scatter (residuals not systematically related to fi
##
## QQ Residuals: if residuals are normally distributed, points should roughly follow the diagonal line.
##
## Scale-Location: random scatter with no clear pattern (equal variance of residuals across all levels o
##
## Residuals vs Leverage: highlights points that are outliers or have high leverage.
```

```
# Coefficient of multiple correlation between the response and predictors
# Value taken from model summary
r_squared <- summary(model)$r.squared
cat("R-squared: ", r_squared, "\n")
```

```
## R-squared:  0.1980115
```

```r
# value of Sales for Price = 150 and City = "Yes"
new_data <- data.frame(Price = 150, City = "Yes")
pred_sales <- predict(model, new_data, interval = "confidence", level = 0.95)
cat("Predicted Sales with 95% confidence interval: \n")
```

```
## Predicted Sales with 95% confidence interval:
```

```r
print(pred_sales)
```

```
##        fit      lwr      upr
## 1 568.9963 522.8507 615.1419
```

```r
# Sales with 95% prediction interval for Price = 150 and City = "Yes"
pred_sales_with_PI <- predict(model, new_data, interval = "prediction", level = 0.95)
cat("Predicted Sales with 95% prediction interval: \n")
```

```
## Predicted Sales with 95% prediction interval:
```

```r
print(pred_sales_with_PI)
```

```
##        fit     lwr      upr
## 1 568.9963 68.4032 1069.589
```

```r
cat("The significant Pearson correlation between Sales and Price in Task 2 (-0.4449507) corresponds to t
```

```
## The significant Pearson correlation between Sales and Price in Task 2 (-0.4449507) corresponds to the
```

TASK 5

```r
ChildCarSeats <- read.csv("data/ChildCarSeats.csv")

# Build logistic regression model
ChildCarSeats$Expensive <- ifelse(ChildCarSeats$Expensive == "Yes", 1, 0)

logit_model <- glm(Expensive ~ Sales, data = ChildCarSeats, family = "binomial")

summary(logit_model)
```

```
##
## Call:
## glm(formula = Expensive ~ Sales, family = "binomial", data = ChildCarSeats)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)  3.3637112  0.4028592    8.35   <2e-16 ***
## Sales       -0.0044786  0.0005207   -8.60   <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 554.48  on 399  degrees of freedom
## Residual deviance: 448.18  on 398  degrees of freedom
## AIC: 452.18
##
## Number of Fisher Scoring iterations: 4
```

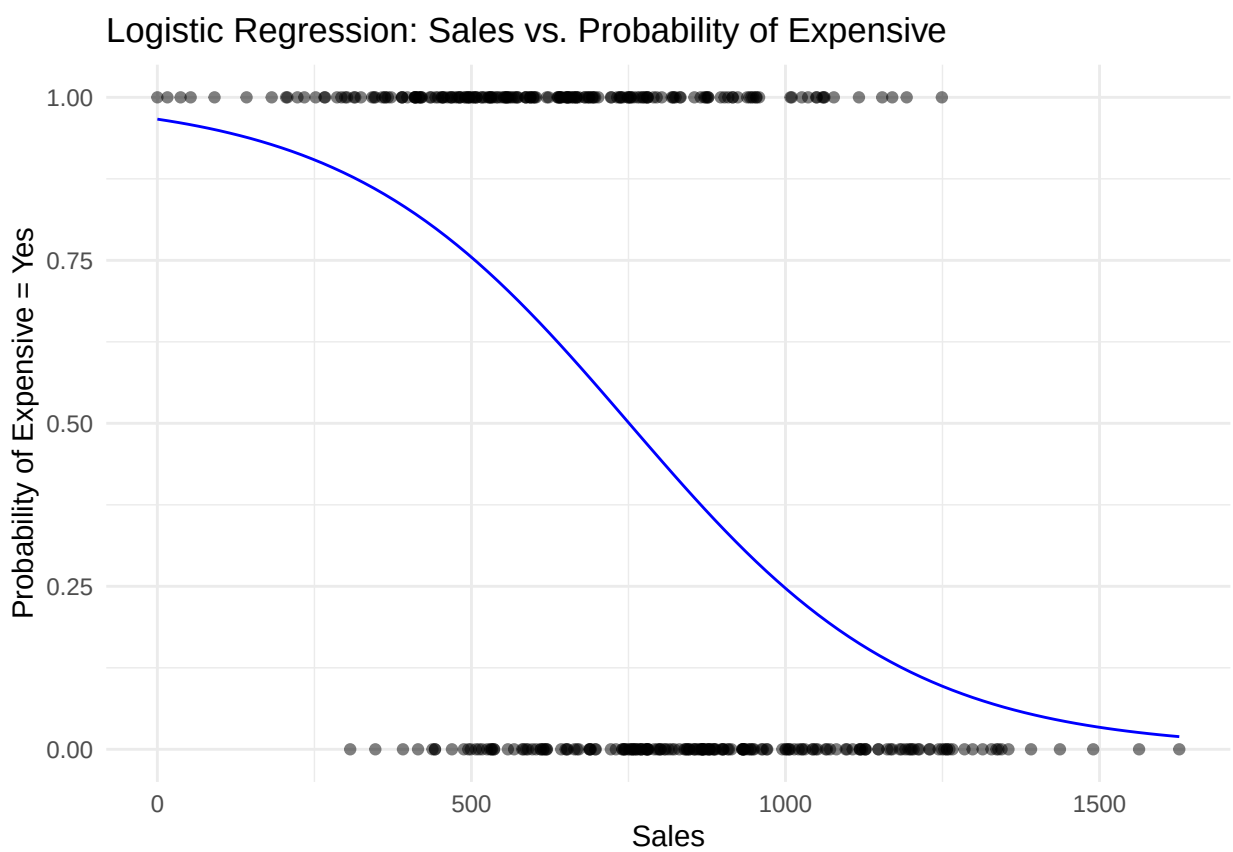
```r
cat("The model indicates that Sales is significantly associated with the probability of an item being la
```

```
## The model indicates that Sales is significantly associated with the probability of an item being labe
```

```r
# Plot logistic regression curve
sales_range <- seq(min(ChildCarSeats$Sales), max(ChildCarSeats$Sales), length.out = 100)

predicted_probabilities <- predict(logit_model, newdata = data.frame(Sales = sales_range), type = "respo

plot_data <- data.frame(Sales = sales_range, Predicted_Prob = predicted_probabilities)
ggplot(ChildCarSeats, aes(x = Sales, y = Expensive)) +
  geom_point(alpha = 0.5) +
  geom_line(data = plot_data, aes(x = Sales, y = Predicted_Prob), color = "blue") +
  labs(x = "Sales", y = "Probability of Expensive = Yes", title = "Logistic Regression: Sales vs. Probal
  theme_minimal()
```



```r
# Calculate probability of Expensive == "Yes" when Sales = 1000
sales_value <- 1000

coefficients <- coef(logit_model)

log_odds <- coefficients[1] + coefficients[2] * sales_value

probability <- 1 / (1 + exp(-log_odds))

se <- sqrt(diag(vcov(logit_model)))
lower_log_odds <- log_odds - 1.96 * se[2] * sales_value
upper_log_odds <- log_odds + 1.96 * se[2] * sales_value
lower_prob <- 1 / (1 + exp(-lower_log_odds))
upper_prob <- 1 / (1 + exp(-upper_log_odds))

cat("Probability of Expensive == 'Yes' when Sales = 1000: ", probability, "\n")
```

```
## Probability of Expensive == 'Yes' when Sales = 1000:  0.2469659
```

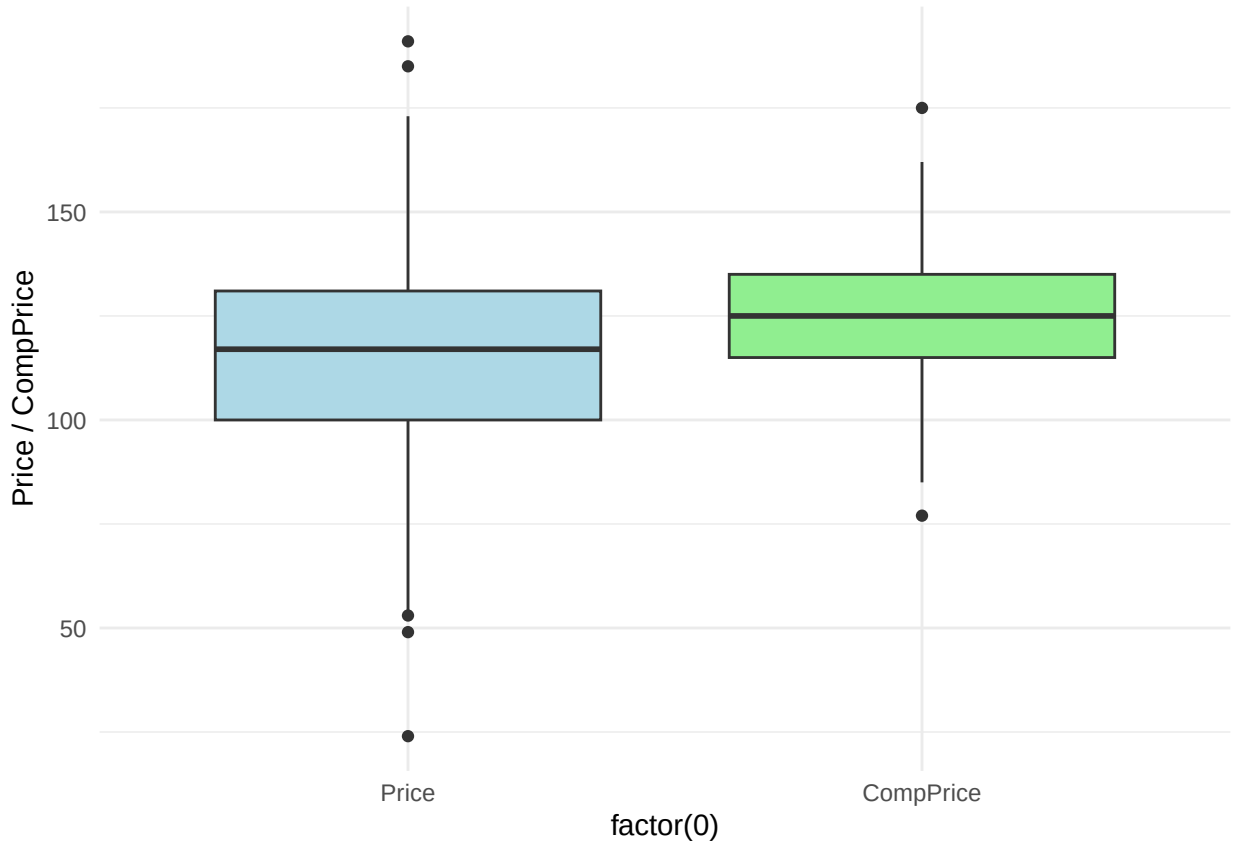
```r
cat("95% Confidence Interval for Probability: [", lower_prob, ", ", upper_prob, "]\n")
```

```
## 95% Confidence Interval for Probability: [ 0.1056929 ,  0.476465 ]
```

TASK 6

```r
ChildCarSeats <- read.csv("data/ChildCarSeats.csv")

# Boxplot for Price and CompPrice
ggplot(ChildCarSeats, aes(x = factor(0), y = Price)) +
  geom_boxplot(fill = "lightblue") +
  geom_boxplot(data = ChildCarSeats, aes(x = factor(1), y = CompPrice), fill = "lightgreen") +
  scale_x_discrete(labels = c("Price", "CompPrice")) +
  ylab("Price / CompPrice") +
  theme_minimal()
```




```r
# Wilcoxon signed-rank test to compare medians of
wilcox_test <- wilcox.test(ChildCarSeats$Price, ChildCarSeats$CompPrice, alternative = "less")
print(wilcox_test)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  ChildCarSeats$Price and ChildCarSeats$CompPrice
## W = 59594, p-value = 2.118e-10
## alternative hypothesis: true location shift is less than 0
```

```r
cat("The p-value is very small (we reject the null hypothesis). Conclusion: the median of Price is sign
```

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## The p-value is very small (we reject the null hypothesis). Conclusion: the median of Price is signif
```